

Ex. 8: Producer Consumer Using Semaphores

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[qaiser@localhost ~]$ cat producerconsumer.c
#include<stdio.h>
#include<stdlib.h>
int mutex=1,full=0,empty=5,x=0;
int wait(int s){
    return s--;
}
int signal(int s){
    return ++s;
}
void producer(){
    mutex=wait(mutex);
    empty=wait(empty);
    full=signal(full);
    x++;
    printf("\nProducer produces item %d",x);
    mutex=signal(mutex);
}
void consumer(){
    mutex=wait(mutex);
    full=wait(full);
    empty=signal(empty);
    printf("\nConsumer consumes item %d",x);
    x--;
    mutex=signal(mutex);
}
int main(){
    int n;
    printf("\n---Producer Consumer Problem---");
    printf("\n1.Producer");
    printf("\n2.Consumer");
    printf("\n3.Exit");
    while(1){
        printf("\nEnter your choice:");
        scanf("%d",&n);
        switch(n){
            case 1:
                if(mutex==1&&empty!=0){
                    producer();
                }
                else{
                    printf("\nBuffer is full!! Producer cannot produce.");
                }
                break;
            case 2:
                if(mutex==1&&full!=0){
                    consumer();
                }
                else{
                    printf("\nBuffer is empty!! Consumer cannot consume.");
                }
                break;
            case 3:
                printf("\Exiting program...");
                printf("\nFinal buffer items produced: %d\n",x);
                exit(0);
            default:
                printf("\nInvalid choice. Try again.");
        }
    }
    return 0;
}
```

```

            case 2:
                if(mutex==1&&full!=0){
                    consumer();
                }
                else{
                    printf("\nBuffer is empty!! Consumer cannot consume.");
                }
                break;
            case 3:
                printf("\Exiting program...");
                printf("\nFinal buffer items produced: %d\n",x);
                exit(0);
            default:
                printf("\nInvalid choice. Try again.");
        }
    }
    return 0;
}
[qaiser@localhost ~]$
```

```
qaiser@localhost:~  
[qaiser@localhost ~]$ gcc producerconsumer.c -o producerconsumer  
[qaiser@localhost ~]$ .\producerconsumer  
bash: ./producerconsumer: command not found..  
[qaiser@localhost ~]$ ./producerconsumer  
  
----Producer Consumer Problem----  
1.Producer  
2.Consumer  
3.Exit  
Enter your choice:1  
  
Producer produces item 1  
Enter your choice:1  
  
Buffer is full!! Producer cannot produce.  
Enter your choice:2  
  
Buffer is empty!! Consumer cannot consume.  
Enter your choice:3  
iting program...  
Final buffer items produced: 1  
[qaiser@localhost ~]$
```